DO NOW — RECALL

Lesson 101 · Multicore CPUs and Parallel Systems

Name: _____ Class: _____ Date: _____

1. What is a core?

2. Define concurrency.

3. What does Amdahl's law illustrate?

4. A 3×4 array of independent cells: how many cells?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A CPU processing unit able to run a hardware thread or more, depending on design.

2. Model answer

Several tasks progress over overlapping time.

3. Model answer

Serial work limits parallel speedup.

4. Model answer

12.

By the end, you can…

• A multicore CPU integrates two or more processing cores, which can execute independent threads concurrently.

• Parallel processing can reduce completion time when tasks can be divided; a serial dependency is a critical-path constraint.

• Shared memory bandwidth, locking, thread scheduling and the serial proportion of a program limit speedup. More cores do not guarantee proportional gains.

Vocabulary — know these terms

multicore — Processor with multiple CPU cores.

parallelism — Simultaneous processing of independent tasks.

critical path — Longest dependent chain.

Amdahl's law — Speedup bounded by serial portion.

NEW LEARNING

Read, annotate and explain

Understand

A multicore CPU integrates two or more processing cores, which can execute independent threads concurrently.

Apply

Parallel processing can reduce completion time when tasks can be divided; a serial dependency is a critical-path constraint.

Evaluate

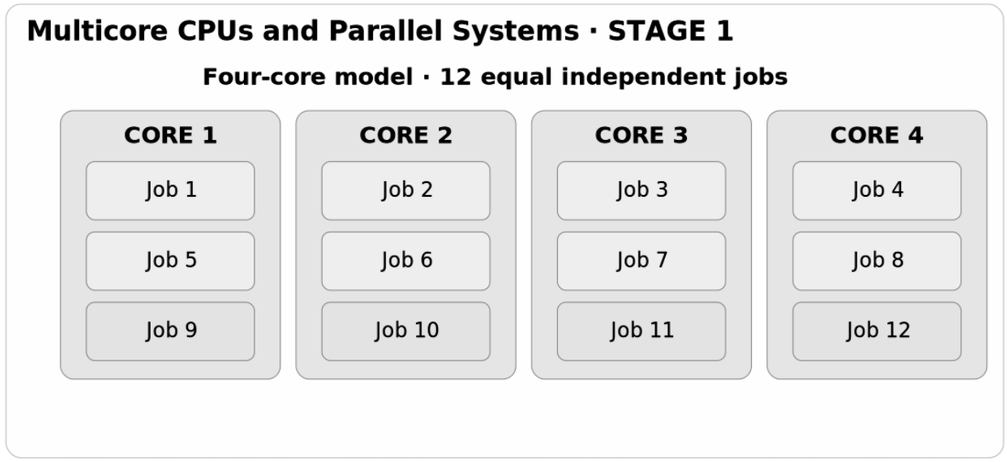
Shared memory bandwidth, locking, thread scheduling and the serial proportion of a program limit speedup. More cores do not guarantee proportional gains.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

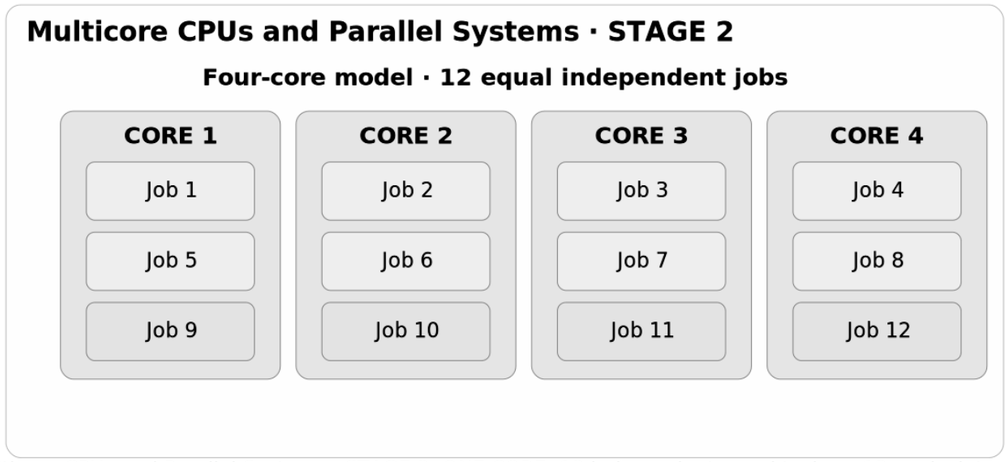
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Multicore CPUs and Parallel Systems · STAGE 1; ONE CORE: 12 independent equal tasks → 12 work slots

New Learning



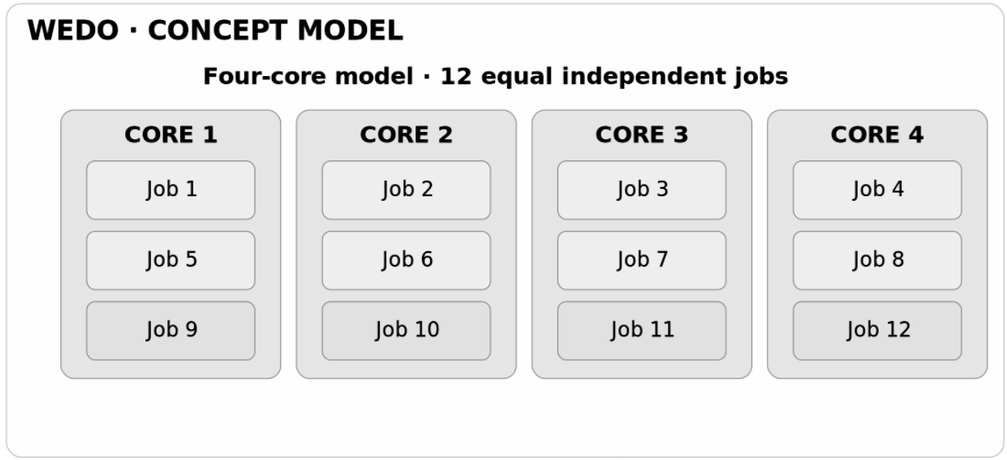
Multicore CPUs and Parallel Systems · STAGE 2; ONE CORE: 12 independent equal tasks → 12 work slots; FOUR CORES: 4 tasks simultaneously → 3 waves

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Twelve equal independent jobs run on four cores. Give ideal waves and one practical caveat.

12 ÷ 4 = 3 waves; shared-memory contention or scheduling overhead can extend time.



WEDO · CONCEPT MODEL; Twelve equal independent jobs run on four cores. Give ideal waves and one practical caveat.: 12 ÷ 4 = 3 waves; shared-memory contention or scheduling overhead can extend time.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

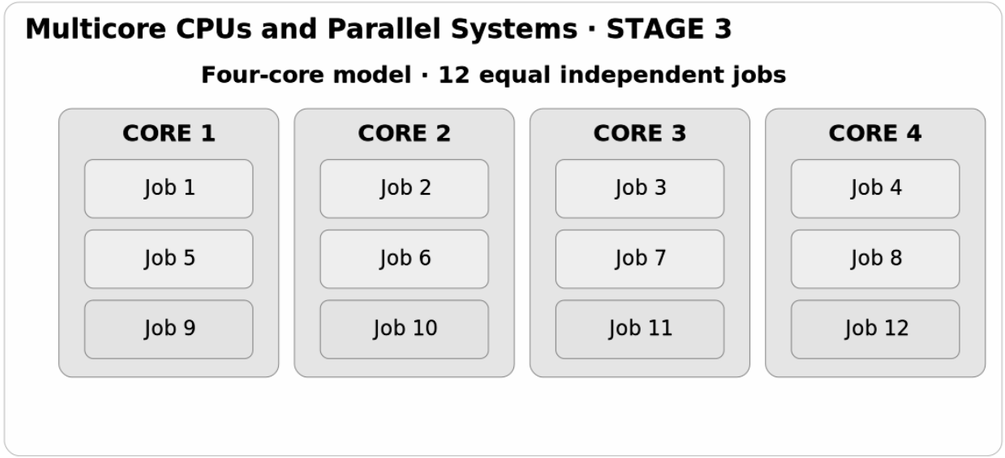
Guided 1 — CHECK

Explain the choice
12 ÷ 4 = 3 waves; shared-memory contention or scheduling overhead can extend time.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Multicore CPUs and Parallel Systems · STAGE 3; ONE CORE: 12 independent equal tasks → 12 work slots; FOUR CORES: 4 tasks simultaneously → 3 waves; REALITY: Shared resources/dependencies limit ideal gains

I DO — WORKED EXAMPLE

Follow the worked method

Eight equal independent jobs are shared across two identical cores. How many ideal waves?

Explain the choice

8 ÷ 2 = 4 equal-duration waves, versus 8 on one core.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Four CPU cores always give exactly four times the performance of one core.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

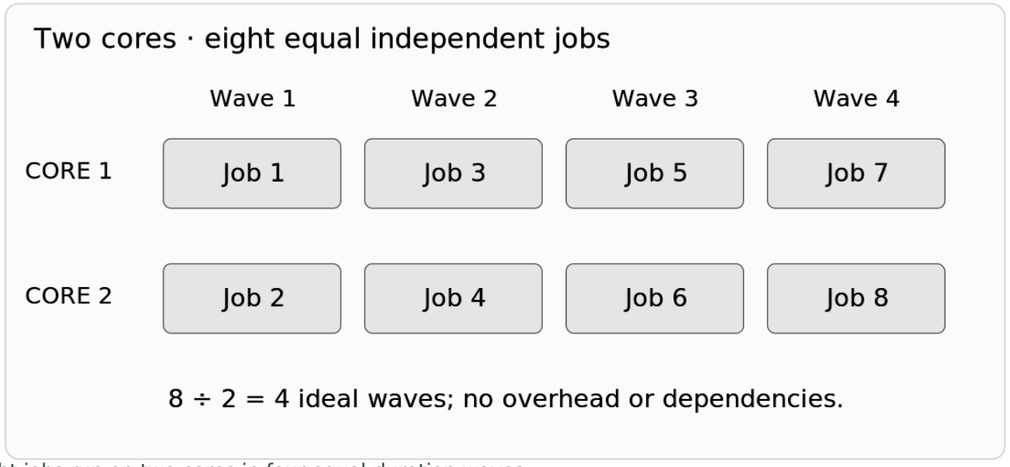
Half a program must run serially; other half parallelised perfectly across four cores. Compute ideal speedup and interpret. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Eight equal independent jobs are shared across two identical cores. How many ideal waves?

8 ÷ 2 = 4 equal-duration waves, versus 8 on one core.



Eight jobs run on two cores in four equal-duration waves

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Twelve equal independent jobs run on four cores. Give ideal waves and one practical caveat.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Twelve equal independent jobs run on three identical cores. Compare ideal waves with one core. [6]

Question 2

Explain two benefits and two limitations of multicore processing for video rendering. [8]